

Deep Sequential Architectures for Financial Time Series Classification

CMPE 679 Deep Learning Project Proposal

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Abstract—Financial time series are nonlinear, noisy, and exhibit complex temporal dependencies. Traditional machine learning models treat observations independently and therefore fail to explicitly capture sequential structure. This project investigates and compares deep sequential neural architectures for short-horizon financial signal classification. Specifically, we evaluate shallow baselines, multilayer perceptrons (MLP), convolutional networks, recurrent architectures (LSTM and GRU), hybrid CNN–RNN models, and attention-based architectures for predicting threshold-defined future directional movements in cryptocurrency data. The objective is to analyze how architectural design influences temporal representation learning and predictive robustness. Performance will be assessed using classification metrics and risk-adjusted financial measures.

I. INTRODUCTION

Financial markets generate sequential data characterized by volatility clustering, regime shifts, heavy-tailed distributions, and non-stationarity. Short-horizon prediction is particularly challenging due to low signal-to-noise ratios and rapidly changing regimes.

While classical machine learning methods can model non-linear feature interactions, they lack structured temporal memory. Deep sequential architectures explicitly model temporal dependencies and may better capture multi-scale financial dynamics.

The central research question is:

How do different sequential deep learning architectures compare in learning temporal representations for short-horizon financial direction prediction?

II. RELATED WORK

A. Short-Horizon Market Predictability

Cryptocurrency markets exhibit high volatility and weak short-term autocorrelation. Jaquart et al. [1] demonstrate that predictive performance strongly depends on horizon length and proper chronological validation. These findings emphasize rigorous time-based splits and careful target construction.

B. Classical Machine Learning Baselines

Logistic regression, support vector machines, random forests, and gradient boosting are commonly used in financial forecasting. Although competitive with engineered features, these models treat observations independently and do not explicitly model sequential structure.

C. MLP and Convolutional Architectures

MLPs approximate nonlinear mappings over fixed windows but lack structured temporal modeling. Convolutional neural networks (CNNs) capture local temporal patterns via shared filters. Temporal Convolutional Networks (TCNs) extend CNNs using dilated causal convolutions to model longer-range dependencies [2].

D. Recurrent Architectures

LSTM introduces gated memory cells to mitigate vanishing gradients and preserve long-term information [3]. GRU simplifies gating with fewer parameters while maintaining performance [4]. Empirical studies show LSTM and GRU are strong baselines for time series forecasting [5].

E. Hybrid CNN–RNN Models

DeepLOB [6] demonstrates that CNN layers can extract hierarchical features which are then modeled sequentially using recurrent networks. This hybrid paradigm motivates CNN–LSTM architectures in financial classification tasks.

F. Event-Based Labeling and Noise Reduction

High-frequency returns are dominated by near-zero movements. Event-based sampling and threshold labeling improve signal separability [7]. This motivates our threshold-based direction labeling approach.

G. Attention and Transformer Architectures

Attention mechanisms allow models to weight relevant timesteps dynamically. Transformers replace recurrence with self-attention [8]. Temporal Fusion Transformer (TFT) integrates attention and gating for interpretable multi-horizon forecasting [9]. These architectures represent higher-capacity baselines for sequential modeling.

III. PROBLEM FORMULATION

Let $x_t \in \mathbb{R}^d$ denote feature vectors constructed from OHLCV data and technical indicators.

We construct sequences:

$$X_t = [x_{t-L+1}, \dots, x_t] \quad (1)$$

Future log-return over horizon H is defined as:

$$r_{t,H} = \log \left(\frac{C_{t+H}}{C_t} \right) \quad (2)$$

Using threshold τ , labels are defined as:

$$y_t = \begin{cases} 1 & \text{if } r_{t,H} > \tau \\ 0 & \text{if } r_{t,H} < -\tau \end{cases} \quad (3)$$

Samples with $|r_{t,H}| \leq \tau$ are discarded to reduce label noise. The model learns:

$$f_{\theta}(X_t) = \hat{y}_t \quad (4)$$

Parameters are optimized using Binary Cross-Entropy:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (5)$$

IV. METHODOLOGY

A. Dataset and Features

We use high-frequency cryptocurrency data (Bitcoin 1-minute). Features include:

- OHLCV
- RSI, MACD
- Bollinger Bands
- Rolling volatility
- Momentum indicators

Data is split chronologically into training, validation, and test sets to prevent look-ahead bias.

B. Architectures Evaluated

- Logistic Regression
- Random Forest / XGBoost
- MLP
- CNN / TCN
- LSTM
- GRU
- CNN-LSTM
- Attention-based models
- Transformer / TFT

C. Regularization and Optimization

- Dropout
- Weight decay
- Early stopping
- Adam optimizer
- Learning rate scheduling

V. EVALUATION

A. Classification Metrics

- Accuracy
- Precision / Recall
- F1-score
- ROC-AUC
- Balanced Accuracy

B. Financial Metrics

- Cumulative return
- Sharpe ratio
- Maximum drawdown
- Win rate

VI. EXPECTED OUTCOMES

We hypothesize:

- Sequential deep architectures outperform shallow baselines.
- CNN-RNN hybrids improve multi-scale pattern extraction.
- Attention-based models improve interpretability and long-horizon modeling.

VII. CONCLUSION

This study systematically compares shallow, recurrent, hybrid, and attention-based architectures for financial time series classification. By applying rigorous evaluation and threshold-based labeling, we aim to provide insight into how deep sequential models learn temporal structure in noisy financial environments.

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